

Pre-Delinquency Intervention Engine

Project Report & Proposal

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1. Executive Summary

The Pre-Delinquency Intervention Engine is a proactive financial risk management system designed to detect early signs of financial distress before a customer misses a payment. By shifting from a reactive "collections-first" approach to a predictive "assistance-first" model, financial institutions can significantly reduce credit losses, lower operational costs, and preserve long-term customer relationships. This report outlines the problem, proposed solution, technology stack, and a practical implementation strategy using our current datasets.

2. Problem Statement

The Reactive Trap: Banks face rising delinquency risks due to economic volatility. Currently, most institutions only intervene *after* a customer has missed a payment (Day 1+ Delinquency). * **Late Action:** By the time a payment is missed, the customer's financial recovery likelihood has already dropped sharply. * **High Cost:** Traditional collections processes are expensive, costing 15-20% of the recovered amount. * **Damaged Relationships:** Aggressive collections tactics strain customer trust and often lead to churn.

The Challenges: 1. **Subtle Signals:** Early indicators of stress (e.g., salary delays, reduced discretionary spend) are scattered across siloed data sources. 2. **Unified View:** Lack of a single, predictive view makes it hard to identify "at-risk" customers who are technically still "current" on their payments. 3. **Generic Outreach:** Existing communication is widespread and generic, often ignored by customers who need specific solutions.

3. Proposed Solution

We propose a **Platform-Agnostic Pre-Delinquency Engine** that analyzes transaction patterns and behavioral signals in real-time to predict the likelihood of default 2-4 weeks in advance.

Core Capabilities

- **Real-Time Pattern Recognition:** Ingests transaction streams to detect anomalies (e.g., stopping of regular salary credits).
- **Predictive Risk Scoring:** Re-calculates a customer's risk score daily based on new activities.
- **Smart Intervention:** Triggers empathy-led outreach (e.g., offering a payment holiday or restructuring) via the preferred channel before the due date.

4. Technology Stack

Our architecture spans a hybrid model leveraging powerful Open Source tools and robust AWS Infrastructure.

Layer	Open Source Component	AWS Service Component	Purpose
Data Ingestion	Apache Kafka	Amazon Kinesis	Real-time transaction streaming & ingestion.

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Data Storage	-	Amazon Redshift / DynamoDB	Historical data warehousing (Redshift) & hot storage (DynamoDB).
Feature Store	Feast	SageMaker Feature Store	Managing real-time ML features (e.g., "avg_spend_30d").
Model Training	XGBoost, LightGBM, PyTorch	Amazon SageMaker	Training classification and sequence models.
Orchestration	Apache Airflow	-	Scheduling data pipelines and retraining jobs.
Model Serving	MLflow, BentoML	SageMaker Endpoints	Deploying models for real-time inference.
Visualization	Plotly, Dash	Amazon QuickSight	Risk dashboards for analysts.
Action	-	Amazon SNS	Triggering notifications/interventions.

5. Implementation Strategy (Using Current Dataset)

We will leverage the existing datasets in `d: /SpaceBar/` to build the MVP of this engine. Here is how we map the theory to our actual data:

A. Dataset Mapping

To detect financial stress, we will correlate data from multiple CSV files currently available:

Customer Profile (Static Risk)

- **Source:** `application_train.csv / application_test.csv`
- **Usage:** Establish baseline risk using Income (AMT_INCOME_TOTAL), Credit Amount (AMT_CREDIT), and Family Status. This tells us *who* the customer is.

Behavioral Signals (Dynamic Risk)

- **Source:** `installments_payments.csv`
- **Signal:** *Payment Hierarchy & Delays.*
 - **Scenario:** A customer usually pays 5 days before the due date. Suddenly, they pay on the due date or 1 day late.
 - **Metric:** DAYS_INSTALLMENT vs DAYS_ENTRY_PAYMENT. Increasing gap = Financial Stress.
- **Source:** `credit_card_balance.csv`
- **Signal:** *Liquidity Crunch.*
 - **Scenario:** Sudden spike in ATM withdrawals (AMT_DRAWINGS_ATM_CURRENT) suggests a desperate need for cash.
 - **Scenario:** Utilization (AMT_BALANCE / AMT_CREDIT_LIMIT_ACTUAL) hitting 90%+ consistently.

- **Source:** POS_CASH_balance.csv
- **Signal:** *slowing repayment.*
 - **Scenario:** Frequent small installment payments instead of full settlement.

B. The "Early Warning" Logic

The engine will monitor specific triggers derived from the data above: 1. **Salary Credit/Income Stability:** (Inferred from inflows in transaction data or application income reliability). 2. **Expense Volatility:** Sudden drop in discretionary spend (inferred from lack of POS transactions in luxury categories). 3. **Debt Service Ratio:** If the AMT_ANNUITY (Loan Payment) > 40% of estimated monthly disposable income.

6. Scope of Work

In-Scope (What we will build)

1. **Data Pipeline:** Ingest application_*.csv, installments_*.csv, and credit_card_*.csv into a unified schema.
2. **Feature Engineering:** Create temporal features (e.g., avg_days_late_last_3_months, cash_withdrawal_ratio).
3. **Prediction Model:** Train an XGBoost Classifier to predict TARGET (Default) but specifically calibrated to prioritize *Recall* (catching as many at-risk cases as possible).
4. **Dashboard:** A Plotly/Streamlit dashboard showing "Top 50 At-Risk Customers" for the week.

Out-of-Scope (Future Phase)

- Integration with live banking core systems (Mainframe).
- Chatbot integration for automated negotiation (Phase 2).

7. Benefits & Impact

- **Reduced Credit Losses:** Intervening 2 weeks early can prevent the default entirely.
- **Operational Efficiency:** Automated detection replaces manual file reviews.
- **Regulatory Goodwill:** Demonstrates "Fair Banking" by offering help before penalties.
- **Customer Retention:** Intervention feels like "support" rather than "harassment."

9. Proof of Concept

To demonstrate the feasibility of this engine using our current data, we have developed a Python script (d:/SpaceBar/poc_risk_features.py) that: 1. **Ingests** a sample of installments_payments.csv. 2. **Calculates** the "Days Past Due" (DPD) for each transaction. 3. **Identifies** customers with an average DPD > 5 days.

Results: In a preliminary run on 10,000 transactions, the script successfully identified **331 customers** exhibiting early signs of financial stress who would be candidates for proactive intervention.

10. Conclusion

By utilizing the detailed transaction history available in our current dataset, we can successfully build a prototype of the Pre-Delinquency Intervention Engine. This system will not only predict risk with high accuracy using XGBoost and Deep Learning but will also provide a framework for actionable, empathetic customer engagement.